

5. Transition Path

Adv. Macro: Heterogeneous Agent Models

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Introduction

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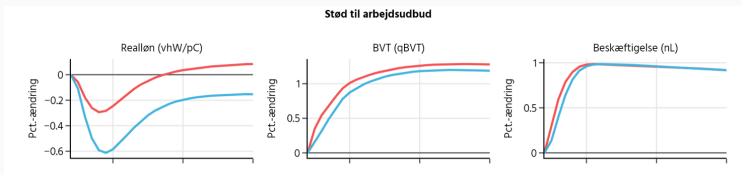
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- **Literature:**
 1. Auclert et al. (2021), »Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models«
 2. Documentation for GEModelTools
 3. Kirkby (2017)

1. Introduction to transitions with the Ramsey model
2. Transition path in HA in partial equilibrium
3. Transition path in HA in general equilibrium: using sequence-space Jacobians
4. First-order approximations of transition paths
5. Fake news algorithm: computing SSJ fast
6. Exercises

Example I

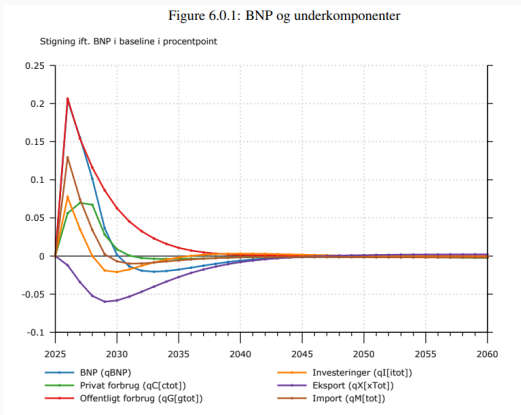
- What do we mean by transition path?
- Permanent shock to labor supply (think increase in retirement age) in the macroeconomic model of the Ministry of Finance:



- Note: Permanent shock, so transition path *between* two different steady states

Example II

- Temporary shock to public spending (i.e. fiscal stimulus during recessions)



- Note: Temporary shock, so model returns to the *same steady state*

Ramsey model

- **Simplified form:**

$$\begin{aligned}u'(C_t^{hh}) &= \beta(1 + F_K(K_t, 1) - \delta)u'(C_{t+1}^{hh}) \\K_t &= (1 - \delta)K_{t-1} + F(K_{t-1}, 1) - C_t^{hh}\end{aligned}$$

- **Production function:** $\Gamma_t K_t^\alpha L_t^{1-\alpha}$

- **Utility function:** $\frac{(C_t^{hh})^{1-\sigma}}{1-\sigma}$

- **Steady state:**

$$\begin{aligned}K_{ss} &= \left(\frac{\left(\frac{1}{\beta} - 1 + \delta \right)}{\Gamma_{ss} \alpha} \right)^{\frac{1}{\alpha-1}} \\C_{ss}^{hh} &= (1 - \delta)K_{ss} + \Gamma_{ss} K_{ss}^\alpha - K_{ss}\end{aligned}$$

Ramsey: As an equation system

$$\begin{bmatrix} r_t^K - \alpha \Gamma_t K_{t-1}^{\alpha-1} L_t^{1-\alpha} \\ w_t - (1-\alpha) \Gamma_t K_{t-1}^{\alpha} L_t^{-\alpha} \\ r_t - (r_t^K - \delta) \\ A_t - K_t \\ A_t^{hh} - ((1+r_t)A_{t-1}^{hh} + w_t L_t^{hh} - C_t^{hh}) \\ C_t^{hh,-\sigma} - \beta(1+r_{t+1})C_{t+1}^{hh,-\sigma} \\ A_t - A_t^{hh} \\ L_t - L_t^{hh} \\ \forall t \in \{0, 1, \dots\}, \text{ given } K_{-1} \end{bmatrix} = 0$$

Remember: Perfect foresight w.r.t aggregate variables

Unknowns: $\{r_t^K, w_t, L_t, K_t, r_t, A_t, C_t^{hh}, A_t^{hh}\}$ for $\forall t \in \{0, 1, \dots\}$

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$$f(x) \approx f(x^i) + f'(x^i)(x - x^i)$$

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$$f(x) \approx f(x^i) + f'(x^i)(x - x^i)$$

- Set $f(x) = 0$ and solve for x to get:

$$x = x^i - \frac{f(x^i)}{f'(x^i)}$$

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- Newton's method: Given initial guess x_0 update guess for x from i to $i + 1$ as:

$$x^{i+1} = x^i - \frac{f(x^i)}{f'(x^i)}$$

- until $|f(x^i)| < \epsilon$

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- How well does it work?
 - If $f(x)$ is linear this update solves $f(x) = 0$ in **1 iteration**
 - If $f(x)$ is non-linear we typically need more iterations, but works well if initial guess is within basin of attraction

Recap: Multivariate Newton's method

- Generalize to vector-valued, multivariate functions $[f_1(x_1, x_2), f_2(x_1, x_2)]' = \mathbf{f}(\mathbf{x})$ with $\mathbf{x} = (x_1, x_2)'$:

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- Where $\mathbf{J}(\mathbf{x}^i)$ is the *Jacobian* of $\mathbf{f}(\mathbf{x})$ w.r.t $\mathbf{x}^i$:

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- If J is expensive to calculate, this is a serious bottleneck
- Broyden's method solves this issue by only calculating J around some initial point.
- Then apply following (linear) update of $f'(x^{i+1})$ at every iteration i :

$$f'(x^{i+1}) = f'(x^i) + \frac{[f(x^{i+1}) - f(x^i)] - f'(x^i)(x^{i+1} - x^i)}{x^{i+1} - x^i}$$

Recap: Broyden's method II

1. Guess $\mathbf{x}^0$ and set $i = 0$
2. Calculate the Jacobian around initial point $\mathbf{J}_0$
3. Calculate $\mathbf{f}^i = \mathbf{f}(\mathbf{x}^i)$.
4. Stop if $\|\mathbf{f}^i\|$ below tolerance ϵ
5. Calculate Jacobian by

$$\mathbf{J}^i = \begin{cases} \mathbf{J}_0 & \text{if } i = 0 \\ \mathbf{J}^{i-1} + \frac{(\mathbf{f}^i - \mathbf{f}^{i-1}) - \mathbf{J}^{i-1}(\mathbf{x}^i - \mathbf{x}^{i-1})}{\|\mathbf{x}^i - \mathbf{x}^{i-1}\|_2^2} (\mathbf{x}^i - \mathbf{x}^{i-1})' & \text{if } i > 0 \end{cases}$$

6. Update guess by $\mathbf{x}^{i+1} = \mathbf{x}^i - (\mathbf{J}^i)^{-1} \mathbf{f}^i$
7. Increment i and return to step 3

- **Go through code**

Back to Ramsey

$$\begin{bmatrix} r_t^K - \alpha \Gamma_t K_{t-1}^{\alpha-1} L_t^{1-\alpha} \\ w_t - (1-\alpha) \Gamma_t K_{t-1}^\alpha L_t^{-\alpha} \\ r_t - (r_t^K - \delta) \\ A_t - K_t \\ A_t^{hh} - ((1+r_t)A_{t-1}^{hh} + w_t L_t^{hh} - C_t^{hh}) \\ C_t^{hh,-\sigma} - \beta(1+r_{t+1})C_{t+1}^{hh,-\sigma} \\ A_t - A_t^{hh} \\ L_t - L_t^{hh} \\ \forall t \in \{0, 1, \dots\}, \text{ given } K_{-1} \end{bmatrix} = 0$$

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- 2 issues:
 - Many unknowns (8 eqs per period)
 - In fact, infinitely many since time is infinite, $T \rightarrow \infty$

Truncated Ramsey, reduced vector form

$$H(K, L, \Gamma, K_{-1}) = \begin{bmatrix} A_t - A_t^{hh} \\ L_t - L_t^{hh} \\ \forall t \in \{0, 1, \dots, T-1\} \end{bmatrix} = 0$$

where $\mathbf{X} = (X_0, X_1, \dots, X_{T-1})$, $A_{-1}^{hh} = K_{-1}$ and

$$r_t^K = \alpha \Gamma_t (K_{t-1}/L_t)^{\alpha-1}$$

$$w_t = (1 - \alpha) \Gamma_t (K_{t-1}/L_t)^\alpha$$

$$A_t = K_t$$

$$r_t = r_t^K - \delta$$

$$C_t^{hh} = (\beta(1 + r_{t+1}))^{-1/\sigma} C_{t+1}^{hh} \text{ (backwards)}$$

$$L_t^{hh} = 1$$

$$A_t^{hh} = (1 + r_t) A_{t-1}^{hh} + w_t L_t^{hh} - C_t^{hh} \text{ (forwards)}$$

Truncation: $T < \infty$ fine when $\Gamma_t = \Gamma_{ss}$ for all $t > \underline{t}$ with $\underline{t} \ll T$

Further reduced

$$H(K, \Gamma, K_{-1}) = [A - A^{hh}] = 0$$

where $\mathbf{X} = (X_0, X_1, \dots, X_{T-1})$, $A_{-1}^{hh} = K_{-1}$ and

$$L_t = L_t^{hh} = 1$$

$$r_t^K = \alpha \Gamma_t(K_{t-1}/L_t)^{\alpha-1}$$

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for $\forall t \in \{0, 1, \dots, T-1\}$

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 - Representing an entire timepath/*sequence* of variables as a function of timepath/*sequence* of other variables

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$$\begin{bmatrix} C_0 & C_1 & C_2 & \dots \end{bmatrix}' = a + b \begin{bmatrix} Y_0 & Y_1 & Y_2 & \dots \end{bmatrix}'$$

$$\Leftrightarrow \mathbf{C} = a + b\mathbf{Y}$$

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- Powerful since it also applies *non-linear*, forward-looking and backwards-looking eqs:

$$C_t = a + b_0 Y_t + b_1 \log Y_{t-4} + b_2 Y_{t+4}^2$$

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- As long as we have the sequence $\mathbf{Y}$ we can calculate $\mathbf{C}$
 - Will leverage this later when working with the HA model

Solution in sequence space

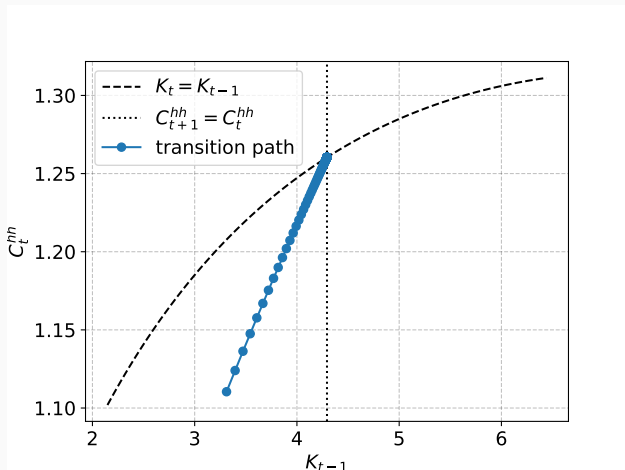
- **Truncation:** $T = 200$ (transition path should have converged to ss by then)
- **Jacobian:** Find $\mathbf{H}_K$ by *numerical differentiation*

$$\mathbf{H}_K = \begin{bmatrix} \frac{\partial(A_0 - A_0^{hh})}{\partial K_0} & \frac{\partial(A_0 - A_0^{hh})}{\partial K_1} & \cdots \\ \frac{\partial(A_1 - A_1^{hh})}{\partial K_0} & \ddots & \ddots \\ \vdots & \ddots & \ddots \end{bmatrix}$$

- **Transition path:** Given Γ and K_{-1} solve $\mathbf{H}(K, \Gamma, K_{-1})$ with non-linear equation system solver (e.g. broyden)
- **Two types of perfect foresight transitions:**
 1. *Transitory:* both the initial and terminal conditions are the steady-state values
 2. *Permanent:* the economy moves from one state to another state (the terminal state must be a stationary one)
- **Notebook:** *01-Ramsey.ipynb*

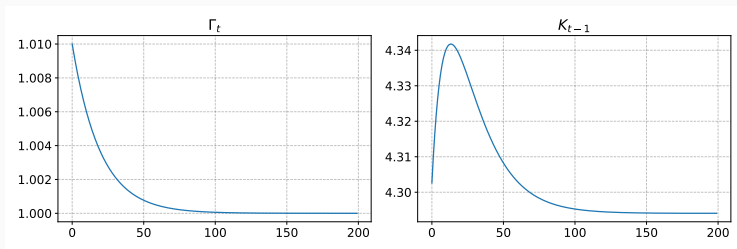
Example 1: permanent from low capital

Initially away from steady state: $K_{-1} = 0.75K_{ss}$



Example 2: transitory following technology shock

Technology shock: $\Gamma_t = \Gamma_{ss} + 0.01 \times \Gamma_{ss} \times 0.95^t$ (i.e AR(1) with $\rho = 0.95$) (exogenous, deterministic)



Terminology: MIT-shock

Transition path in PE

Household model in a transition

Recall the household block in the HANC model

$$v_0(z_{i0}, a_{i,-1}) = \max_{\{c_{it}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$$

s.t.

$$\ell_{it} = z_{it}$$

$$a_{it} = (1 + r_t)a_{it-1} + w_t \ell_{it} - c_{it}$$

$$\log z_{it+1} = \rho_z \log z_{it} + \psi_{it+1}, \quad \psi_{it} \sim \mathcal{N}(\mu_\psi, \sigma_\psi), \quad \mathbb{E}[z_{it}] = 1$$

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Perfect foresight, initial and terminal conditions

Important assumptions:

1. **Perfect foresight:** from $t = 0$, households know the future path of $\{r_t, w_t\}_{t=0}^{\infty}$
2. **Truncation:** the model converges to a stationary state after $t \geq T$, T large
3. **Initial conditions:** we compute the transition from a given distribution D_0 that we already know
4. **Terminal condition:** we compute a transition towards some stationary state where we know the value function (or its derivative)

Impulse responses: backward and forward step

Our goal is to compute a sequence of impulse responses

$$A_t^{hh}(\{r_\tau, w_\tau\}_{\tau=0}^{T-1}) = \int a_t(a, z) dD_t(a, z) \quad \forall t \in \{0, 1, \dots, T-1\}$$

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1. **Backward step:** using the terminal condition on the value function, and going back in time, obtain the policy functions $a_t(a, z)$ and $c_t(a, z)$
2. **Forward step:** using the initial condition on the distribution, and going forward in time, simulate the distribution over time $D_t(a, z)$

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3. An **initial condition** on the distribution

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1. An **exogenous path** of $\{r_t, w_t\}_{t=0}^T$
2. A **terminal condition** on the value function (or its derivative)
 $V_T^a(a, z)$
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We can then obtain the aggregate values of the household as usual by computing $A_t^{hh} = \int a_t(a, z) dD_t(a, z)$. This is the **impulse response**!

Let's code!

Transition path in GE

Equation system

The model can be written as an **equation system**

$$\begin{bmatrix} r_t^K - F_K(K_{t-1}, L_t) \\ w_t - F_L(K_{t-1}, L_t) \\ r_t - (r_t^K - \delta) \\ A_t - K_t \\ \underline{D}_t - \Pi'_z \underline{D}_t \\ \underline{D}_{t+1} - \Lambda'_t \underline{D}_t \\ A_t^{hh} - A_t \\ L_t^{hh} - L_t \\ \forall t \in \{0, 1, \dots\}, \text{ given } \underline{D}_0 \end{bmatrix} = 0$$

where $\{\Gamma_t\}_{t \geq 0}$ is a given technology path and $K_{-1} = \int a_{-1} d\underline{D}_0$

Remember: Policies and choice transitions depend on prices

1. Policy function: $x_t^* = x^* \left(\{r_\tau, w_\tau\}_{\tau \geq t} \right)$ and
 $X_t^{hh} = \sum_i x_{it}^* D_{it} = \mathbf{x}_t^{*'} \underline{D}_t$
2. Choice transition: $\Lambda_t = \Lambda \left(\{r_\tau, w_\tau\}_{\tau \geq t} \right)$

Transition path - close to verbal definition

For a given $\underline{D}_0$ and a path $\{\Gamma_t\}$

1. Quantities $\{K_t\}$ and $\{L_t\}$,
2. prices $\{r_t\}$ and $\{w_t\}$,
3. the distributions $\{D_t\}$ over z_t and a_{t-1}
4. and the policy functions $\{a_t^*\}$, $\{\ell_t^*\}$ and $\{c_t^*\}$

are such that in all periods

1. Firms maximize profits (prices)
2. Household maximize expected utility (policy functions)
3. D_t is implied by simulating the household problem forwards from $\underline{D}_0$
4. Mutual fund balance sheet is satisfied
5. The capital market clears
6. The labor market clears
7. The goods market clears

Reduce size of equation system

- In the equation system above we have many **unknowns** and many **equations**
- Makes finding the solution with Broyden's method slow since the **Jacobian is large**
 - With truncation T and N equations/unknowns $\mathbf{J}$ has size $(T \times N, T \times N)$
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 $\Rightarrow$ Expensive to calculate
- We can typically **exploit model structure** to reduce size of system
 - Did this earlier for Ramsey
 - Now more formally

Truncated, reduced vector form

$$H(K, L, \Gamma, \underline{D}_0) = \left[\begin{array}{c} A_t^{hh} - A_t \\ L_t^{hh} - L_t \\ \forall t \in \{0, 1, \dots, T-1\} \end{array} \right] = 0$$

where $\mathbf{X} = (X_0, X_1, \dots, X_{T-1})$, $K_{-1} = \int a_{-1} d\underline{D}_0$ and

$$r_t^K = \alpha \Gamma_t (K_{t-1}/L_t)^{\alpha-1}$$

$$w_t = (1 - \alpha) \Gamma_t (K_{t-1}/L_t)^\alpha$$

$$r_t = r_t^K - \delta$$

$$A_t = K_t$$

$$\underline{D}_t = \Pi'_z \underline{D}_t$$

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$$A_t^{hh} = \mathbf{a}_t^{*'} \underline{D}_t$$

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$$\forall t \in \{0, 1, \dots, T-1\}$$

Truncation: $T < \infty$ fine when $\Gamma_t = \Gamma_{ss}$ for all $t > \underline{t}$ with $\underline{t} \ll T$

DAG - Directed Acyclic Graph

- **Orange square:** Shocks (exogenous)
- **Blue square:** Unknowns (endogenous)
- **Green circles:** Blocks (with variables and targets inside)



- This DAG implies: Exo. input + guess $\Rightarrow$ Firm block $\Rightarrow$ Mutual fund $\Rightarrow$ HHs $\Rightarrow$ Residuals

Further reduction

$$H(\mathbf{K}, \Gamma, \underline{\mathbf{D}}_0) = \begin{bmatrix} A_t^{hh}(\mathbf{w}(\mathbf{K}), \mathbf{r}(\mathbf{K})) - K_t \\ \forall t \in \{0, 1, \dots, T-1\} \end{bmatrix} = \mathbf{0}$$

where $\mathbf{X} = (X_0, X_1, \dots, X_{T-1})$, $K_{-1} = \int a_{-1} d\underline{\mathbf{D}}_0$ and

$$L_t = 1$$

$$r_t^K = \alpha \Gamma_t (K_{t-1}/L_t)^{\alpha-1}$$

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Solve with Broyden

- As with standard Ramsey model from before we have:
 - Equation system with T equations (H)
 - And T unknowns (K)
- If we can calculate the jacobian of H w.r.t K we can solve with Broyden's method as before

How to compute Jacobian?

- How do we compute the Jacobian of the residuals H w.r.t unknowns K ?
 - Before: Compute Jacobian of entire model using num. diff
 - **Now:** Use DAG structure + chain rule

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- *Example*. Represent model in block form:

$$\mathbf{w}, \mathbf{r}^K = \text{Firm}(\mathbf{K}), \quad \mathbf{A}, \mathbf{r} = \text{MutFund}(\mathbf{K}, \mathbf{r}^K)$$

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- Collapsing the previous equations, we write the asset-market clearing condition as

$$\mathbf{H} = \mathbf{A}^{hh}(\mathbf{w}(\mathbf{K}), \mathbf{r}(\mathbf{K})) - \mathbf{K}$$

What is a Jacobian

Let $\mathcal{J}^{y,x}$ be Jacobian of y w.r.t x . Then:

$$\mathbf{H}_K = \mathcal{J}^{A^{hh},r} \mathcal{J}^{r,K} + \mathcal{J}^{A^{hh},w} \mathcal{J}^{w,K} - \mathbf{I}$$

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Interpretation: row t of column s gives us the savings change at t in response to a shock on r at s . Not just a computational tool, also a lot of economic intuition behind it!

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 - Cheap, and can often be vectorized
 - What about HH Jacobians $\mathcal{J}^{A_{hh},r}, \mathcal{J}^{A_{hh},w}$?
 - Need to compute T impulse responses!

Bottleneck: How do we find the Jacobian?

- **Naive approach:** For each input i into HH block $i \in \{r, w\}$
 - For each $s \in \{0, 1, \dots, T-1\}$
 1. Shock input i in period s by small amount Δ
 2. Solve household problem backwards along transition path
 3. Simulate households forward along transition path
 4. Calculate column s , row t of jacobian as $\frac{\partial \mathcal{J}_t^{A_{hh}, i}}{\partial i_s} = \frac{A_t^{hh} - A_{ss}^{hh}}{\Delta}$ for all t

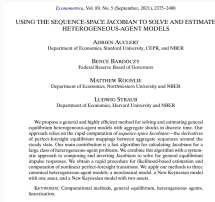
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- **Solution: Fake news algorithm** - only need T steps! (later)



- Conditional on being able to compute HH jacobian efficiently we can compute **transition path** through following steps:
 1. Compute stationary state of model
 2. Formulate transition path as DAG
 - Reduce number of unknowns and residual equations
 - Not essential, but often good idea
 3. Compute Jacobian of residuals H w.r.t unknowns K
 4. Formulate shock (e.g. TFP increases by 1% for 4 years)
 5. Use Broyden's method to solve for transition path

Let's code!

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3. To find $\mathbf{K}$, we use Newton's method, which require a **Sequence Space Jacobian**

Linear transitions

- Unknowns: U

Reminder of model class

- Unknowns: $\mathbf{U}$
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- In deterministic, perfect foresight model (MIT shocks), solve $H(\mathbf{U}, \mathbf{Z}) = 0$ by
 1. Calculating the Jacobian of H w.r.t $\mathbf{U}$ around s.s.
 2. Use Newton's method to find non-linear transition given $\mathbf{Z}$ $\Rightarrow$ But we have abstracted from real aggregate uncertainty

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 - Implies that aggregate shocks are not random process, but rather *MIT shocks*
 - Interpretation of MIT shocks generally hard to reconcile with business cycles

Stochastic vs deterministic models

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- Assume $x_t = \rho^x x_{t-1} + \epsilon_t^x$ with $\mathbb{E}_t^x = 0$. Period 0 solution in deterministic/perfect foresight model:

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- Same result! Aggregate uncertainty **does not matter to first-order** when linearizing w.r.t aggregate shock

- Solve for IRFs for unknowns using first-order approximation

$$H(\mathbf{U}, \mathbf{Z}) = 0 \Rightarrow H_U d\mathbf{U} + H_Z d\mathbf{Z} = 0 \Leftrightarrow d\mathbf{U} = \underbrace{-H_U^{-1} H_Z}_{=G_U} d\mathbf{Z}$$

- We can find H_U and H_Z as before, by chaining Jacobians along the DAG
- Limitations:
 - Imprecise for large shocks
 - Imprecise in models with aggregate non-linearities
 - No real aggregate uncertainty (precautionary savings w.r.t. aggregate shocks, etc)

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You want to compute an IRF facing a TFP shock in HANC.

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Write a first-order approximation

$$H_{\mathbf{K}} d\mathbf{K} + H_{\Gamma} d\Gamma = 0 \Leftrightarrow d\mathbf{K} = -H_{\mathbf{K}}^{-1} H_{\Gamma} d\Gamma$$

Linearized IRFs: example from HANC

You want to compute an IRF facing a TFP shock in HANC.

Recall that

$$H(\mathbf{K}, \mathbf{\Gamma}) = \mathbf{A}(r(\mathbf{K}, \mathbf{\Gamma}), w(\mathbf{K}, \mathbf{\Gamma})) - \mathbf{K}$$

The derivative with respect to $\mathbf{K}$ is

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This gives us the endogenous response $d\mathbf{K}$ for any shock $d\mathbf{\Gamma}$ by simply computing the product of two matrices!

Simulating a time-series using the linearized solution

Idea: in a linear model, the response to many shocks is the sum of the responses to each shock alone

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- **Adding up:** today's dX_t is the sum of the effects of all past innovations

Simulating a time-series: an example

$\text{IRF}^X = (1, 0.5, 0.25, 0, \dots)$, innovations $\epsilon_0 = 1$, $\epsilon_1 = -2$, $\epsilon_2 = 0.5$

	$t = 0$	$t = 1$	$t = 2$	$t = 3$
effect of $\epsilon_0 = 1$	1	0.5	0.25	0
effect of $\epsilon_1 = -2$		-2	-1	-0.5
effect of $\epsilon_2 = 0.5$			0.5	0.25
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In general, adding up the effects of all past innovations:

$$dX_t = \sum_{s=0}^{T-1} \text{IRF}_s^X \epsilon_{t-s}$$

s is the *age* of a shock: ϵ_{t-s} hit s periods ago, and IRF_s^X is how much a shock that old still moves X today

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Why is this allowed? Each ϵ_t is a surprise when it hits, like an MIT shock, and to first order the response does not depend on when it hits

Let's code!

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- In deterministic model $\sigma_{x,t}^2 = 0$ - not true in stochastic model!
 - Models deviate once we go beyond 1st order approximation (linearization)
- Still extremely useful though - we may solve deterministic models to first-order and interpret as models with aggregate uncertainty
 - How do we linearize models numerically?

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 2. Linearize and solve model to get IRF of $\{dC_t\}_{t=0}^T = d\mathbf{C}$ w.r.t $\{dG_t\}$
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- Same principle with more shocks

- **Covariances:**

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- **Covariance decomposition:**

$$\frac{\text{contribution from one shock}}{\text{contributions from all shocks}} = \frac{\sigma_j^2 \sum_{s=0}^{T-1-k} dC_s^j dY_{s+k}^j}{\sum_{i \in \mathcal{Z}} \sigma_i^2 \sum_{s=0}^{T-1-k} dC_s^i dY_{s+k}^i}$$

Solving HA model with aggregate risk (advanced)

- To solve models with aggregate risk we need to write them in *state-space* form instead of *sequence-space*
 - Think of HA household problem - that is always in state-space form
 - Endogenous variables c_t, a_t as function of current states a_{t-1}, z_t

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$$\begin{bmatrix} \mathbf{u}_t \\ \mathbf{z}_t \end{bmatrix} = \mathcal{M} \left(\begin{bmatrix} \mathbf{u}_{t-1} \\ \mathbf{z}_{t-1} \end{bmatrix}, \epsilon_t \right)$$

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- In standard NK model: no backward looking eqs. so number of state variables = Number of shocks

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$$v(\mathbf{D}_t, \Gamma_t, z_{it}, a_{it-1}) = \max_{a_{it}, c_{it}} u(c_{it}) + \beta \mathbb{E}_t [v(\mathbf{D}_{t+1}, \Gamma_{t+1}, z_{it+1}, a_{it})]$$

s.t.

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$$a_{it} + c_{it} = (1 + r_t) a_{it-1} + w_t z_{it}$$

$$\log z_{it+1} = \rho_z \log z_{it} + \psi_{it+1}, \quad \psi_{it} \sim \mathcal{N}(\mu_\psi, \sigma_\psi), \quad \mathbb{E}[z_{it}] = 1$$

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- $\mathbf{D}_t$ is a state variable $\Rightarrow$ Massive state space

- **State-space approach with linearization:** Ahn et al. (2018); Bayer and Luetticke (2020); Bhandari et al. (2023); Bilal (2023)

Con:

1. Harder to implement
2. Valuable to be able to interpret Jacobians

Pro:

1. Easier path to 2nd and higher order approximations
- **Global solution:** The distribution of households is a state variable for each household $\Rightarrow$ *explosion in complexity*
 1. Original: Krusell and Smith (1997, 1998); Algan et al. (2014);
 2. Deep learning: Fernández-Villaverde et al. (2021); Maliar et al. (2021); Han et al. (2021); Kase et al. (2022); Azinovic et al. (2022); Gu et al. (2023); Chen et al. (2023)
 - **Discrete aggregate risk:** Lin and Peruffo (2023)

Fake News Algorithm

- Household block:

$$\mathbf{Y}^{hh} = hh(\mathbf{X}^{hh})$$

- e.g. $\mathbf{Y}^{hh} = C^{hh}, A^{hh}$ and $\mathbf{X}^{hh} = w, r$

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- Shock at time $s = 0$, solve + simulate HH block for T periods
- Repeat until $s = T - 1$
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- **Next slides:** *Sketch of much faster approach*

- Note that aggregate is (matrix) product of individual policy function $\mathbf{y}_t$ and distribution $\mathbf{D}_t$.
- Linearize (first-order Taylor) around ss:

$$\begin{aligned}\mathbf{Y}^{hh} &= (\mathbf{y}'_t) \mathbf{D}_t \\ \Rightarrow \frac{d\mathbf{Y}^{hh}}{d\mathbf{X}^{hh}} &= \left(\frac{d\mathbf{y}_t}{d\mathbf{X}^{hh}} \right)' \mathbf{D}_{ss} + (\mathbf{y}'_{ss}) \frac{d\mathbf{D}_t}{d\mathbf{X}^{hh}}\end{aligned}$$

- What can we say about policy function term $d\mathbf{y}_t$?

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Then:

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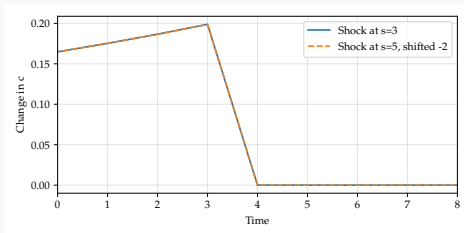
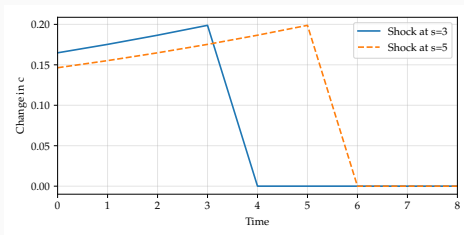
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- **Implication:** We need to only do a single backwards iteration to a shock at $s = T - 1$.
 - Can then construct change in policy function $d\mathbf{y}_t^s/d\mathbf{X}^{hh}$ for different s by shifting policy function around

Numerical illustration

Graphically. Response of c_t to income shock at $s = 3, 5$



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Implementation

Let's say you want to compute $\frac{d\mathbf{A}^{hh}}{d\mathbf{w}}$

Algorithm:

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Let's code!

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- GEModelTools does all of this »under the hood« when you compute HH Jacobians
 - You just tell GEModelTools the inputs and outputs of the household block
 - Entire algorithm is automated

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4. Sequence Space Jacobians are costly to compute with 'brute-force', but fake-news algorithm is fast!

Exercises

Same model. Your choice of τ_{ss} . New questions:

1. **Define the transition path.**
2. **Plot the DAG**
3. **What do the Jacobians look like?**
4. **Find the transition path for $G_t = G_{ss} + 0.01G_{ss}0.95^t$**
5. **What explains household savings behavior?**
6. **What happens to consumption inequality?**

Answers available at this [link](#)

Recap

Recap: consumption-saving

- The buffer-stock model: income risk and a borrowing constraint give precautionary saving, a concave consumption function, and high MPCs at low wealth
- The solution is a pair of policy functions $a^*(z, a_{-1})$ and $c^*(z, a_{-1})$
- EGM (backward step): grid on the choice a_t , invert the Euler equation, endogenous grid from the budget constraint, interpolate back, impose the constraint
- Histogram method (forward step): the distribution is mass on the grid, moved by the productivity transition Π'_z and the savings lottery Λ'
- Aggregates are sums of policies over the distribution,
$$A^{hh} = \sum \mathbf{D} a^*$$

Recap: stationary equilibrium

- In general equilibrium, prices are endogenous and set by market clearing
- HANC = buffer-stock households + firms + mutual fund + markets
- The definition of the stationary equilibrium
- Solving it is a one-dimensional root-finding problem: guess K , get (r, w) , solve backward, simulate forward, return $A_{ss}^{hh} - K$
- Bisection is safe but slow, Newton is fast but needs f' and can diverge; brentq combines the two
- Precautionary saving pushes r_{ss} below $1/\beta - 1$

Recap: transition path

- Transitory vs permanent transitions, and their initial and terminal conditions
- Nonlinear vs linear solutions
- Perfect foresight (MIT shocks) and truncation at T
- The equilibrium in sequence space: $\mathbf{H}(\mathbf{K}, \mathbf{\Gamma}) = \mathbf{A}^{hh}(\mathbf{r}, \mathbf{w}) - \mathbf{K} = \mathbf{0}$
- The household impulse response: backward from $\underline{v}_{a,T}$, forward from $\underline{\mathbf{D}}_0$
- Newton's method needs the sequence-space Jacobian, built with the chain rule along the DAG
- Linear solution: $d\mathbf{U} = \mathbf{G} d\mathbf{Z}$, and simulation as a sum of past shocks

Recap: additional topics

- To first order, the IRF to an MIT shock is the IRF of the model with aggregate risk (certainty equivalence)
- Fake news algorithm: policies only depend on the distance to the shock, so one backward pass gives the whole Jacobian
- With perceived aggregate risk, the distribution becomes a state variable: Krusell-Smith, state-space perturbation, deep learning